Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics

2.3 – Series Circuits

Unit 2: Electricity & Energy

The question: *in a loop with several bulbs, is the current the same everywhere — and does each bulb get the full push from the cell, or does the push get shared out?* Today's circuits all have one thing in common: **one loop, no junctions.** That single fact decides both rules below.

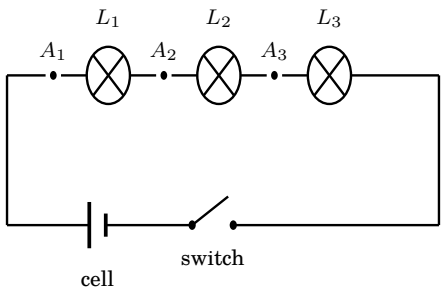
Learning intentions

All students will state and apply the **current rule** and the **voltage rule** for series circuits.
Most students will explain what adding another bulb does to the others, without a number.
Some students will calculate a missing voltage using both rules together.

Part 1 — The current rule

In a series circuit, the current has **only one path** round the loop — there is nowhere else for it to go. So:

The current is the SAME at every point in a series circuit.



1) **Predict** the readings at A_2 and A_3, if A_1 reads **0.4 A**.

Build it

Build the three-bulb circuit and place your ammeter at A_1, then A_2, then A_3 in turn. Record all three.

$A_1 =$ _____ A $A_2 =$ _____ A $A_3 =$ _____ A

Check: all three readings should be the same, within the accuracy of your meter.

Name: _____ **Date:** _____

Part 2 — The voltage rule

The cell gives every bit of charge a fixed push — a fixed voltage. That push gets **used up** crossing each bulb in turn, so:

> **The voltages across the components ADD UP to the cell's voltage.**

2) **Calculate** the voltage across L_2, if the cell is **6 V** and L_1 reads **2.5 V**.

Build it

Measure the voltage across the cell, then across L_1 and L_2 only (ignore L_3 for this check).

Cell = _____ V L_1 = _____ V L_2 = _____ V

Check: $L_1 + L_2$ should add up close to the cell's voltage — not exactly, because a real circuit loses a little voltage in the wires and the meters themselves.

Name: _____ **Date:** _____

Most students — adding another bulb

3) **Explain** what happens to the brightness of **every** bulb in the three-bulb circuit if you add a fourth bulb into the same loop. You do not need a number — explain using the current rule.

Some students — using both rules together

4) **Calculate** the voltage across L_3 in a series circuit with a **9 V** cell, where $L_1 = 3\,\text{V}$ and $L_2 = 4\,\text{V}$.

Part 3 — On your own

A series circuit has a cell and three bulbs. The ammeter reads **0.2 A**. The cell is **12 V**. L_1 reads **5 V** and L_2 reads **3 V**.

5) **Find** the current through L_3, and the voltage across L_3. Show which rule you used for each.

HOMEWORK

Old-style fairy lights are wired in series — one loop, all the bulbs. If **one** bulb in the string burns out and its filament breaks, explain what happens to the **rest** of the string, and why — using what you learned about series circuits today.